GGG-V-790D
August 27, 1985
SUPERSEDING
GGG-V-790C
September 27, 1972

FEDERAL SPECIFICATION

VULCANIZERS, HOT PATCH AND PLATE, INNER TUBE

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers vulcanizers for applying tube patches and rubber-covered-base valve stems to inner tubes.

1.2 Classification.

1.2.1 Types and styles. Vulcanizers covered by this specification shall be of the following types and styles, as specified (see 6.2):

Type I - Deleted.
Type II - Hot patch vulcanizer (flammable heat units).
  Style A - Tube patch and tube valve.
  Style B - Tube patch.
Type III - Plate vulcanizer (electrically heated).
  Style A - Deleted.
  Style B - Single-tube patch and tube valve.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Standards

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under "General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	-	Varnish, Moisture- and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-V-45554	-	Vulcanizing Equipment, Including Related Items, Tire and Tube Rebuild and Repair, Preparation For Delivery of

Military Standards

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulation:

Occupational Health and Safety Administration (OSHA) 29 CFR 1910.211-222
General Industrial Safety and Health Standards

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

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ASTM D3951-82 - Standard Practice for Commercial Packaging
ASTM E380-84 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Fire Protection Association (NFPA)

NFPA 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one or more vulcanizers for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The vulcanizers shall be new and of the manufacturer's latest design which shall meet the requirements specified herein. The vulcanizers shall be designed to accomplish tube repairs without damage to the tubes. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the vulcanizer or any vulcanizer part or component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the vulcanizers. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Safety devices. Safety devices shall be provided for parts that present safety hazards to personnel. Safety devices shall not interfere with the operation of the vulcanizer or with the operator's performance.

3.2.2.1 Occupational safety standards. The product shall comply with OSHA standards set forth in 29 CFR 1910.211-222 that are applicable to vulcanizers.

3.2.3 Interchangeability. Parts shall be designed and manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

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3.2.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. Parts used in fabrication of the vulcanizers shall be new and free of defects. Part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Screws, pins, bolts, and similar parts shall be installed with means to prevent loss of tightness and adjustment. Parts subject to adjustment or removal shall not be swaged, peened, staked, or otherwise deformed. Construction of the vulcanizers shall be rugged and sturdy, and indicative of equipment produced by manufacturers normally involved in producing this type of equipment.

3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

3.4.4 Protective finishes. All exposed metal (see 6.5.1) and working surfaces, except those made of corrosion-resistant materials, shall be coated or covered with a protective finish that provides protection against corrosion.

3.5 Performance and product characteristics.

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3.5.1 Type II. Type II vulcanizers shall be either wall or bench-mounted vulcanizers which incorporate facilities for holding and applying self-vulcanizing patch units (flammable heat units) up to at least three inches in diameter. The style A vulcanizer shall be suitable for applying tube patches and rubber-covered-base valve stems onto inner tubes of sizes up to, and including, 14.00-24. The style B vulcanizer shall be suitable for applying tube patches onto those same size inner tubes.

3.5.2 Type III. Type III vulcanizers shall be electrically-heated, bench or floor mounted, which incorporate facilities for holding and curing tube patches and rubber-covered-base valve stems onto inner tubes of sizes up to and including, 14.00-24. The style B vulcanizer shall be suitable for repairing one inner tube at one time. The curing head on the style B vulcanizer shall be a minimum of 12 inches long by four inches wide. The time required for bringing the electrically-heated vulcanizers to vulcanizing temperature prior to performing the repair operations shall not exceed 30 minutes.

3.6 Components.

3.6.1 Clamping units. Clamping units shall meet the requirements for the particular type and style specified in 3.6.1.1 and 3.6.1.2.

3.6.1.1 Type II, styles A and B. The type II vulcanizers shall incorporate a hand-operated, quick-acting clamp with the required spider(s) and pressure pad to apply the repairs specified in 3.5.1. The clamp shall hold the repairs in position without loosening during the vulcanizing operation.

3.6.1.2 Type III. The type III vulcanizers shall be provided with hand-operated, adjustable screw or quick-acting clamps, including the required spiders or pads, for use with the plates specified in 3.6.2. The clamps shall hold the tube and patch in position without loosening during the vulcanizing process. The style B vulcanizer shall incorporate a single clamp.

3.6.2 Plates (type III). The type III vulcanizer shall be equipped with plates for tube patch and valve stem repairs on inner tubes as required in 3.5.2.

3.6.3 Electrical equipment (type III). Electrical equipment used on type III vulcanizers shall conform to NFPA 70.

3.6.3.1 Heating elements. Electric heating elements shall be of the resistance type and compatible with the requirements of 3.5.2. Electric heating elements shall be insulated to prevent current leakage and shall distribute heat evenly throughout the vulcanizing plates. Unless otherwise specified (see 6.2), heating elements shall be operated from 115 volts, 60 hertz, single-phase current.

3.6.3.2 Automatic control. Electrically-heated vulcanizers shall be equipped with an automatic temperature control which shall be adjustable to any setting from +287° to +312° Fahrenheit (F). The automatic control shall maintain the set temperature within $\pm$5°F at any setting.

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3.6.3.3 Wiring. Wiring used either within, or adjacent to, heat zones of the electrically-heated vulcanizers shall be protected from the heat by being suitably insulated. Exposed wiring shall be of multiple-conductor type encased in a protective sheath. Wiring shall be of a size that shall carry the maximum sustained current requirements of the vulcanizer without damage.

3.6.4 Temperature indicator (type III). Type III vulcanizers shall be furnished with a temperature indicator to indicate when the plates are at vulcanizing temperature.

3.6.5 Accessory equipment. A roughener rasp or a hand-operated wire buffer shall be furnished with each vulcanizer. The tool shall be attached with a chain, or other suitable means, to the vulcanizer to prevent loss.

3.7 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.8 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.9 Product identification. For military activities, the vulcanizers shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number. For civil agencies, the vulcanizers shall be marked in a plain and permanent manner with the manufacturer's name or trademark in such a way that the source of manufacture can be readily identified.

3.10 Workmanship. Standards of workmanship shall assure that the vulcanizers shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Temperature control and accuracy test (see 4.6.2).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. Unless otherwise specified (see 6.2), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL (Percent Defective)
Product examination (4.5)	100 percent inspection	-
Performance test (4.6.1)	II	1.0
Temperature control and accuracy test (4.6.2)	II	1.0
Packaging inspection (4.7)	S-4	4.0

4.5 Product examination. The vulcanizers shall be visually and manually examined to determine conformance with the requirements of 3.2 through 3.3, 3.4 through 3.4.4, and 3.6.3 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the vulcanizers have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.5.1 Evidence of compliance with OSHA standards. Evidence of compliance with applicable safety and health requirements shall be reviewed (see 3.2.2.1 and 6.4).

4.6 Tests.

4.6.1 Performance test. Vulcanizers shall be operated in accordance with the manufacturer's instructions in order to conduct the applicable performance test in 4.6.1.1 and 4.6.1.2.

4.6.1.1 Type II. Type II vulcanizers shall be used to apply self-vulcanizing (flammable heat unit) repairs as follows:

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Style A - A three inch diameter tube patch and a valve stem repair on an inner tube.
Style B - A three inch diameter tube patch on an inner tube.

Repairs shall be examined for imperfections, and tubes shall be examined for damage, to determine compliance with 3.5.1 and 3.6.1.1.

4.6.1.2 Type III. The type III vulcanizers shall be connected to an applicable source of power and allowed to preheat. Upon completion of the preheating period, the vulcanizers shall be used to apply a three inch diameter tube patch and a valve stem repair on an inner tube. The vulcanizers shall be observed during operation. Repairs shall be examined for imperfections, and tubes shall be examined for damage, to determine compliance with the following:

(a) Functional requirements (see 3.5.2).
(b) Preheating time (see 3.5.2).
(c) Clamping unit operation (see 3.6.1.2).
(d) Automatic control requirements (see 3.6.3.2).

4.6.2 Temperature control and accuracy test (type III). Electrically heated vulcanizers shall be connected to an applicable source of power and allowed to preheat to vulcanizing temperature (see 3.5.2). Temperature of the plates shall then be monitored for a period of at least ten minutes, using instrumentation which has an accuracy of 0.5°F, to determine compliance with 3.6.3.2.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing.

5.1.1 Department of Defense.

5.1.1.1 Military. When specified (see 6.2), requirements for Level A preservation and Level A and B packing shall be in accordance with MIL-V-45554.

5.1.1.2 Commercial. When specified (see 6.2), preservation and packing shall be in accordance with ASTM D3951-82.

5.1.2 Civil agencies. When specified (see 6.2), unit packaging shall provide adequate protection against damage during multiple shipment, handling and storage. Packaged units shall be packed in a manner to afford adequate protection against damage during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation. The supplier may use his standard practice when it meets these requirements.

5.2 Marking.

5.2.1 Department of Defense.

5.2.1.1 Military. When specified (see 6.2), requirements for marking shall be in accordance with MIL-STD-129.

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5.2.1.2 Commercial. When specified (see 6.2), requirements for marking shall be in accordance with ASTM D3951-82.

5.2.2 Civil agencies. When specified (see 6.2), requirements for marking for shipment for civil agencies shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. Type II vulcanizers use flammable heat units (consumable) to acquire their vulcanizing heat and are intended for use in field shops to apply self-vulcanizing patches and valve stem repairs to inner tubes (depending upon the style selected). Type III vulcanizers are electrically heated and are intended for use in base and field shops to apply tube patches and valve stem repairs to inner tubes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and style required (see 1.2.1).
(c) First article when required (see 3.1).
(d) Heating element power source (voltage) if different (see 3.6.3.1).
(e) Painting and finishing if different (see 3.8).
(f) Product identification if different (see 3.9).
(g) First article inspection if different (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Sampling if different (see 4.4.1).
(j) Level of preservation and packing (see 5.1.1.1, 5.1.1.2, and 5.1.2).
(k) Type of marking required (see 5.2.1.1, 5.2.1.2, and 5.2.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 OSHA standards. Products which meet all requirements of this specification comply with the OSHA standards published prior to the date of this specification and identified in Section 3 (see 3.2.2.1).

6.5 Definitions.

6.5.1 Exposed metal. Exposed metal is defined as metal exposed to atmospheric conditions, solvents, chemicals, and heat (see 3.4.4).

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MILITARY INTERESTS:

Custodians
Army - AL
Navy - YD

Review Activities
Army - ME
DLA - CS

User Activities
Army - AT
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army - AL

DoD Project 4910-0572

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GGG-V-790D	Vulcanizers, Hot Patch and Plate, Inner Tube

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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